THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 9 · SET 02

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. Which of the following correctly matches a cell organelle with its function?

(a) Ribosome – digestion of worn-out organelles

(b) Lysosome – synthesis of proteins

(c) Golgi apparatus – packaging and dispatch of materials

(d) Mitochondria – photosynthesis

Ans: (c) Golgi apparatus packages and dispatches materials; the others are mismatched.

2. A wilted lettuce leaf becomes crisp when it is kept in tap water. This is because:

(a) minerals diffuse out of the cells

(b) the cell walls dissolve in water

(c) water leaves the cells by osmosis

(d) water enters the cells by osmosis and makes them turgid

Ans: (d) Tap water is hypotonic to the cell sap, so water enters by osmosis and the cells become turgid.

3. Leaves and stems of a canna plant bend easily without breaking. The tissue that provides this flexibility is:

(a) parenchyma (b) collenchyma (c) sclerenchyma (d) xylem

Ans: (b) Collenchyma gives flexibility along with mechanical support.

4. Blood is classified as a connective tissue because it:

(a) flows inside blood vessels

(b) has a fluid matrix (plasma) and connects different parts of the body by transporting substances

(c) contains haemoglobin

(d) contains cells

Ans: (b) Connective tissue has cells in a matrix; blood's matrix is plasma and it links organs by transport.

5. Which substances released from old refrigerators, air-conditioners and aerosol sprays damage the ozone layer?

(a) Carbon monoxide (b) Chlorofluorocarbons (CFCs)

(c) Methane (d) Nitrogen

Ans: (b) CFCs release chlorine atoms in the stratosphere that break down ozone.

6. Mushrooms and moulds are placed in kingdom Fungi because they:

(a) make food by photosynthesis

(b) are prokaryotes

(c) have chitin in their cell wall and absorb food from dead organic matter

(d) are motile and ingest food

Ans: (c) Fungi are heterotrophic, saprotrophic, with chitinous cell walls.

7. Which part of a flower develops into the fruit after fertilisation?

(a) Ovule (b) Ovary (c) Stigma (d) Anther

Ans: (b) The ovary becomes the fruit and ovules become seeds.

8. **Assertion (A):** Tendons are strong but have very little elasticity.
Reason (R): Tendons attach muscles to bones.

Ans: (b) Both are true, but attachment of muscle to bone does not explain their low elasticity.

9. **Assertion (A):** An increase in the amount of carbon dioxide in the atmosphere raises the temperature of the Earth.
Reason (R): Carbon dioxide absorbs outgoing infrared radiation and re-radiates part of it back to the Earth.

Ans: (a) R is the mechanism behind the greenhouse effect described in A.

10. Write any two differences between self-pollination and cross-pollination.

Self: pollen transferred to stigma of the same flower/plant; no genetic variation [1]. Cross: pollen transferred to stigma of another flower of a different plant; variation produced; agent needed. [1]

O R

Name two agents of pollination and give one plant pollinated by each.

Wind – maize/wheat; insects/bees – sunflower/mustard; birds/water also accepted. [1 + 1]

11. Name the kingdom to which each of the following belongs and write one distinguishing feature: (i) *Amoeba* (ii) mushroom (iii) mango tree.

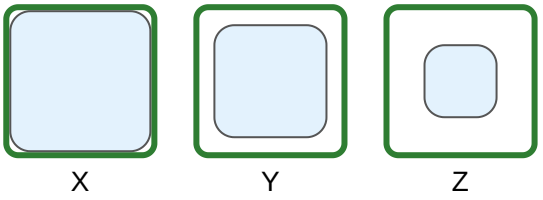
(i) Protista – unicellular eukaryote [1] (ii) Fungi – heterotrophic, chitin cell wall [1] (iii) Plantae – multicellular autotroph with cellulose wall [1]

O R

Differentiate between monocots and dicots on any three points, with one example each.

Seed leaves: one/two; Leaf venation: parallel/reticulate; Roots: fibrous/tap root; flower parts in 3s/4–5s. Example: wheat/maize; gram/pea. [Any three differences 3 × 1; examples as support]

12. The figure shows the same plant cell kept in three different solutions X, Y and Z. State whether each solution is hypotonic, isotonic or hypertonic to the cell sap. Also name the phenomenon observed in the cell Z.



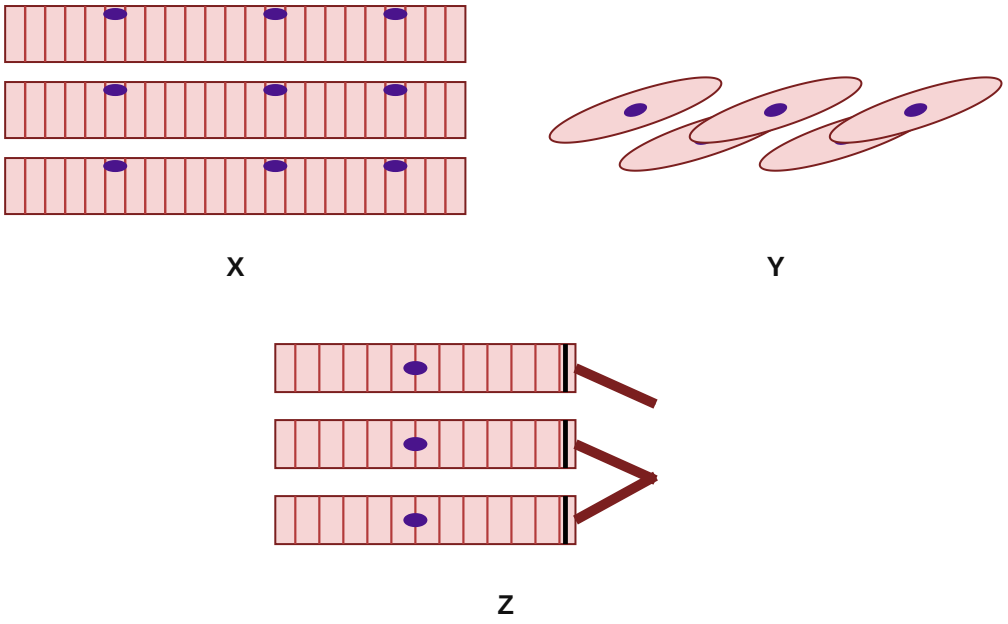
X – hypotonic [1]; Y – isotonic [1]; Z – hypertonic, plasmolysis [1]

13. Fill in the blanks of the table:

Tissue	Location	Function
Phloem	Vascular bundles of plants	(a) _____
(b) _____	Wall of the heart	Rhythmic contraction to pump blood
Areolar tissue	(c) _____	Supports internal organs and helps in repair

(a) Transport of food (sugars) from leaves to other parts [1] (b) Cardiac muscle [1] (c) Between skin and muscles / around blood vessels and nerves / in the spaces inside organs [1]

14. The figure shows three types of muscle fibres X, Y and Z. Identify each and name one place in the human body where it is found.



X Y Z

> X – skeletal (striated) muscle: limbs/biceps [1]; Y – smooth (unstriated) muscle: wall of stomach/intestine/blood vessels [1]; Z – cardiac muscle: heart [1]

15. While playing football, Ravi twisted his ankle badly. The doctor said that a ligament was torn and advised rest. Ravi's teacher later explained that ligaments help the joints work, and that the human body has different types of joints such as those at the knee, shoulder and neck.

(a) What is a ligament? State its function.

> A strong, flexible connective tissue joining one bone to another; it holds bones in position at a joint. [1]

(b) Name the type of joint present at the knee.

> Hinge joint. [1]

(c) Differentiate between a tendon and a ligament (any two points).

> Tendon connects muscle to bone, inelastic, strong; ligament connects bone to bone, elastic/flexible. [1 + 1]

OR

(c) Name the type of joint at the (i) shoulder and (ii) neck (skull–vertebra). State one movement each permits.

> (i) Ball-and-socket – rotation in all directions [1] (ii) Pivot joint – side to side/nodding movement [1]

16. (i) State the three main postulates of the cell theory.
(ii) Name the scientist who first observed cells and the material he used.

> (i) All living organisms are made of cells; the cell is the basic unit of life; all cells arise from pre-existing cells. [3] (ii) Robert Hooke [1], thin slice of cork [1]

OR

Distinguish between a plant cell and an animal cell on any five points.

> Cell wall present/absent; chloroplasts present/absent; large central vacuole/small or absent; centrioles absent/present; shape fixed/irregular, etc. [5 × 1]

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. A mixture of acetone (boiling point 56 °C) and water (boiling point 100 °C) can best be separated by:

(a) filtration (b) simple distillation
(c) using a separating funnel (d) centrifugation

> **Ans: (b)** Miscible liquids with sufficiently different boiling points are separated by distillation.

18. The percentage by mass of oxygen in calcium carbonate, $CaCO_3$ (Ca = 40, C = 12, O = 16), is:

(a) 12 % (b) 40 % (c) 48 % (d) 60 %

> **Ans: (c)** Formula mass = 100 u; oxygen = 48 u → 48 %.

19. Which of the following best describes a scientific model?

(a) An exact full-size copy of a system
(b) A simplified representation used to explain or predict the behaviour of a system
(c) A final, never-changing truth
(d) A guess that is never tested

> **Ans: (b)** Models simplify reality and are revised with new evidence.

20. **Assertion (A):** Colloidal particles can be seen with the naked eye.
Reason (R): Colloids scatter a beam of light passing through them.

> **Ans: (d)** A is false – colloidal particles are too small to be seen. R is true (Tyndall effect).

21. Why is a mixture of salt and water separated by evaporation, whereas a mixture of two miscible liquids is separated by distillation?

> Salt is non-volatile, so water alone evaporates leaving the salt [1]; miscible liquids differ in boiling points, so the lower boiling one vaporises first and is condensed. [1]

22. 4 g of hydrogen combines completely with 32 g of oxygen to form water. State the law and find the mass of water formed.

> Law of conservation of mass – mass of products = mass of reactants [1]; water = 4 + 32 = 36 g [1]

23. Write the electronic configuration of sodium (atomic number 11). State its valency and give a reason.

> 2, 8, 1 [1]; valency 1 – it can lose its single outer electron to attain a stable octet. [1]

24. 25 g of sugar is dissolved in 225 g of water. (i) Calculate the % m/m of the solution. (ii) How much water must be added to 100 g of this solution to reduce its concentration to 5 %?

> (i) 25/250 × 100 = 10 % [1] (ii) Sugar in 100 g = 10 g; new mass = 10/0.05 = 200 g [1]; water to add = 100 g [1]

25. How does fractional distillation differ from simple distillation? Give one example where it is used.

> Simple: liquids with large bp difference; Fractional: uses a fractionating column for liquids with close bps [2]. Example: separation of gases of air / crude oil fractions. [1]

OR

Why is crystallisation preferred to evaporation to dryness for purifying a solid? Give one example.

> Evaporation to dryness may decompose/ char the solid and leaves soluble impurities in the residue [1½]; crystallisation gives pure, regular crystals as impurities stay in the solution [1]. Example: copper sulphate/alum/sugar [½]

26. A student prepared three sugar solutions of different concentrations at room temperature, as shown in the table.

Beaker	Sugar (g)	Water (g)
P	5	95
Q	10	90
R	20	80

(Solubility of sugar at room temperature ≈ 200 g per 100 g water.)

(a) Which beaker has the highest concentration (% m/m)?

R (20 %). [1]

(b) Calculate the % m/m of beaker Q.

10/100 × 100 = 10 %. [1]

(c) Is the solution in beaker R saturated or unsaturated? Justify.

Unsaturated [1]; 20 g per 80 g water is far below the solubility limit (≈ 160 g per 80 g). [1]

OR

(c) How much sugar is present in 50 g of the solution of beaker R?

20 % of 50 g [1] = 10 g [1]

27. A kerosene–water mixture and an acetone–water mixture are to be separated. Name the technique for each, give the principle, and describe the procedure for the first.

Kerosene–water: separating funnel (immiscible liquids differ in density) [1½]; acetone–water: distillation (difference in bp) [1½]. Procedure: pour mixture into funnel, allow layers to settle, open stopcock to drain the lower (water) layer, close it when the interface reaches the stopcock; collect kerosene. [2]

OR

(a) Differentiate between a solution, a suspension and a colloid on the basis of particle size, Tyndall effect and stability. (b) Why does milk not settle on standing?

(a) Solution: <1 nm, no Tyndall, stable; Colloid: 1–100 nm, shows Tyndall, stable; Suspension: >100 nm, shows Tyndall, settles. [3 × 1] (b) Colloidal particles are very small and keep moving (Brownian motion), hence do not settle. [2]

SECTION C — PHYSICS — 12 questions · 23 marks

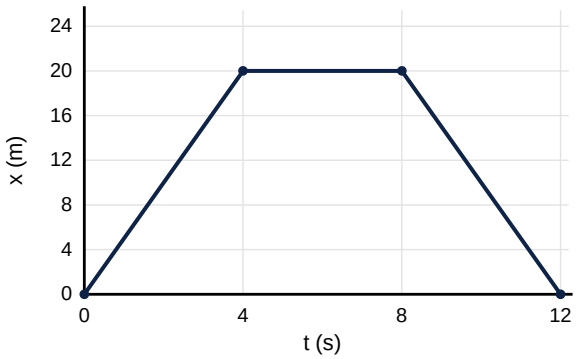
Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A cyclist covers the first half of a distance at 10 m s^{-1} and the second half at 15 m s^{-1}. His average speed for the whole journey is:

(a) 12.5 m s^{-1} (b) 12 m s^{-1} (c) 13 m s^{-1} (d) 25 m s^{-1}

Ans: (b) $v = 2v_1v_2/(v_1+v_2) = 2\times10\times15/25 = 12$ m s^{-1}.

29. The displacement–time graph of a body is shown. Its velocity between t = 0 and t = 4 s is:



(a) 5 m s^{-1} (b) 20 m s^{-1} (c) 0 (d) 80 m s^{-1}

> **Ans: (a)** Slope = 20 m / 4 s = 5 m s^{-1}.

30. A 10 kg box is pushed by a horizontal force of 30 N but does not move. The force of friction acting on it is:

(a) zero (b) 10 N (c) 30 N (d) 100 N

> **Ans: (c)** The box is in equilibrium, so static friction equals the applied force, 30 N.

31. A truck of mass 2000 kg moving at 5 m s^{-1} and a car of mass 1000 kg moving at 10 m s^{-1} are compared. Their momenta are:

(a) greater for the truck (b) greater for the car

(c) equal (d) cannot be compared

> **Ans: (c)** Both: 2000×5 = 1000×10 = 10 000 kg m s^{-1}.

32. A body of mass 1 kg has a kinetic energy of 50 J. Its speed is:

(a) 5 m s^{-1} (b) 10 m s^{-1} (c) 25 m s^{-1} (d) 50 m s^{-1}

> **Ans: (b)** $v = \sqrt{(2E/m)} = \sqrt{100} = 10$ m s^{-1}.

33. An echo is heard 2 s after a clap. If the speed of sound in air is 340 m s^{-1}, the distance of the reflecting wall is:

(a) 170 m (b) 340 m (c) 680 m (d) 1360 m

> **Ans: (b)** d = vt/2 = 340 × 2 / 2 = 340 m.

34. **Assertion (A):** The area under a speed–time graph gives the acceleration of the body.
Reason (R): The slope of a velocity–time graph gives the acceleration of the body.

> **Ans: (d)** A is false – the area gives the distance. R is true.

35. A constant force of 12 N acts on a 3 kg body initially at rest for 4 s. Find (i) its acceleration and (ii) its velocity at the end of 4 s.

> (i) a = F/m = 12/3 = 4 m s^{-2} [1] (ii) v = at = 16 m s^{-1} [1]

36. A person standing in a bus falls backward when the bus suddenly starts. Name the law that explains this and explain.

> Newton's first law (inertia of rest) [1]; the feet move with the bus but the upper body tends to remain at rest. [1]

OR

Why is it harder to stop a loaded truck than an empty one moving at the same speed?

> Loaded truck has greater mass, hence greater momentum (inertia) [1]; so a larger force/longer time is needed to stop it. [1]

37. A ship sends an ultrasonic pulse of frequency 30 kHz towards the sea floor and receives the echo after 0.8 s. If the speed of sound in sea water is 1500 m s^{-1}, calculate (i) the depth of the sea and (ii) the wavelength of the pulse.

> (i) d = vt/2 = 1500 × 0.8/2 = 600 m [1½] (ii) λ = v/f = 1500/30000 = 0.05 m [1½]

38. On Sports Day, Rohit tried to push a 40 kg wooden box across the playground. The maximum static friction between the box and the ground is 120 N and the kinetic friction is 80 N.

(a) What is the force of friction on the box when Rohit pushes with 100 N and the box does not move?

> 100 N (static friction equals the applied force). [1]

(b) Rohit now pushes with 160 N and the box starts sliding. Calculate its acceleration.

> a = (160 − 80)/40 = 2 m s⁻². [1]

(c) Why is kinetic friction less than maximum static friction?

> Once the box moves, the surface irregularities do not get enough time to interlock [1]; hence less force is needed to keep it moving. [1]

OR

(c) Write two ways in which friction between two surfaces can be reduced.

> Polishing/smoothening the surfaces; lubrication (oil/grease); using ball bearings/wheels. [Any two × 1]

39. A stone is dropped from the top of a tower and reaches the ground after 4 s ($g = 10\ m\ s^{-2}$). Find (i) its velocity on hitting the ground, (ii) the height of the tower, (iii) the distance fallen in the last second, (iv) the average velocity during the fall and (v) the momentum on hitting the ground if its mass is 0.5 kg.

> (i) v = gt = 40 m s⁻¹ [1] (ii) h = ½gt² = 80 m [1] (iii) 80 − ½×10×9 = 35 m [1] (iv) 80/4 = 20 m s⁻¹ [1] (v) 0.5 × 40 = 20 kg m s⁻¹ [1]

OR

A car of mass 1000 kg moving at $20\ m\ s^{-1}$ is brought to rest in 10 s by uniform braking. Calculate (i) its initial momentum, (ii) the retardation, (iii) the braking force, (iv) the distance travelled before stopping and (v) the work done by the braking force.

> (i) 20 000 kg m s⁻¹ [1] (ii) 2 m s⁻² [1] (iii) 1000 × 2 = 2000 N [1] (iv) s = ½ × 20 × 10 = 100 m [1] (v) W = 2000 × 100 = 2 × 10⁵ J [1]